AZIM PREMJI UNIVERSITY, BHOPAL

First Semester 2026–2027
Tutorial Sheet–9

Course No. MTH-113 Course title: Introduction to Quantitative Thinking

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(1) Compare the following pairs of numbers using $>$ or $<$.
(a) $0 \square -8$; (b) $-1 \square -15$; (c) $5 \square -5$;
(d) $11 \square 15$; (e) $0 \square 6$; (f) $-20 \square 2$

(2) Write all the integers between the given pairs (write them in the increasing order.)

(a) 0 and -7 (b) -4 and -4 (c) -8 and -15 (d) -11 and 10 (e) -30 and -23

(3) Suppose your friend says following:
(a) Every positive integer is larger than every negative integer.
(b) Zero is less than every positive integer.
(c) Zero is larger than every negative integer.
(d) Zero is neither a negative integer nor a positive integer.
(e) Farther a number from zero on the right, larger is its value.
(f) Farther a number from zero on the left, smaller is its value.
Do you agree with your friend? Give examples.

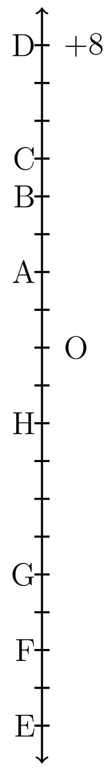
(4) By looking at the number line, answer the following questions:
(a) Which integers lie between -8 and -2?
(b) Which is the largest integer and the smallest integer among them?

(5) Write opposite of the following:
(a) Increase in weight; (b) 30 KM north; (c) 326BC;
(d) Loss of Rs 700; (e) 100 m above sea level.

(6) Draw a number line and answer the following:
(a) Which number will we reach if we move 4 numbers to the right of -2?
(b) Which number will we reach if we move 5 numbers to the left of 1.
(c) If we are at -8 on the number line, in which direction should we move to reach -13?
(d) If we are at -6 on the number line, in which direction should we move to reach -1?

(7) (a) Write four negative integers greater than -20.
(b) Write four negative integers less than -10.

(8) Adjacent figure is a vertical number line, representing integers. Observe it and locate the following points:
(a) If point D is $+8$, then which point is -8?
(b) Is point G a negative integer or a positive integer?
(c) Write integers for points B and E.
(d) Which point marked on this number line has the least value?
(e) Arrange all the points in decreasing order of value.



(9) Find the solution of the following:
(a) (+7) + (-11)
(b) (-13) + (+10)
(c) (-7) + (+9)

(d) (+10) + (-5)
(e) (– 217) + (– 100)

(10) Find the sum
(a) (– 7) + (– 9) + 4 + 16
(b) (37) + (– 2) + (– 65) + (– 8)

(11) Fill in the blanks with $>, <$ or $=$ sign.
(a) (–3) + (–6) $\square$ (–3)–(–6)
(b) (–21)–(–10) $\square$ (–31) + (–11)
(c) (–25)–(–42) $\square$ (–42)–(–25)

(12) Fill in the blanks
(a) $(-8) + \square = 0$
(b) $13 + \square = 0$
(c) $\square - 15 = -10$
(d) $(-4) + \square = -12$

(13) Find
(a) (– 7) – 8 – (– 25)
(b) (– 13) + 32 – 8 – 1
(c) (– 7) + (– 8) + (– 90)
(d) 50 – (– 40) – (– 2)

(14) Between 1 and 20 (including both of them) write
(a) all even numbers
(b) all odd numbers
(c) all prime numbers
(d) all composite numbers

(15) Find the prime factorization for: (a) 30 (b) 60 (c) 72 (d) 84 (e) 210

(16) (a) List the common prime factors of 256 and 156.
(b) Shreya has two tarps, 84 inches and 96 inches wide. She wants to cut them into strips of equal width as wide as possible. What is the greatest width she can cut?

(17) (a) Compute Highest Common Factors (HCF) of following pairs:
(a) $24, 32$ (b) $15, 225$ (c) $28, 35$ (d) $1024, 1020$
(b) Compute Least Common Multiples (LCM) of following pairs:
(a) $3, 4$ (b) $12, 16$ (c) $21, 15$ (d) $13, 169$

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